

GOOD SAMPLE LAB

Lab II, Problem 4: Projectile Motion and Mass

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Introduction

A group of medieval warfare enthusiasts is planning a reenactment and intends to build a trebuchet. If the reenactment is to be safe and realistic, the motion of the projectiles it launches must be well understood. The acceleration of the projectile is constant in time, as confirmed by a previous experiment. This experiment sought to understand the mass dependence of that constant acceleration. To do so, the projectiles were modeled using balls; the trebuchet, using an experimenter's arm. The hypothesis that the acceleration is mass-independent was confirmed.

Prediction

It is hypothesized that the acceleration of an object in projectile motion is mass-independent; this is depicted graphically in Figure 1.

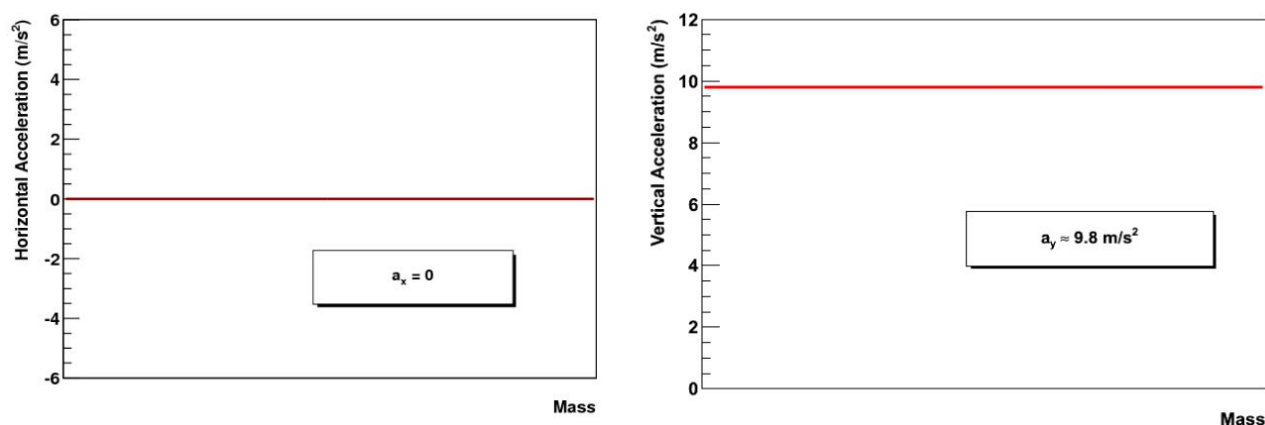


Figure 1: Horizontal and vertical components of acceleration of a projectile near Earth's surface.

The acceleration of all objects moving ballistically near the surface of Earth is downward and of a magnitude given by local g , approximately 9.8 m/s^2 , i.e. is constant with respect to mass. Mathematically,

$$\frac{\Delta \vec{a}}{\Delta m} = \vec{0}$$

This is an assumption of our theory of kinematics.

Procedure

Spherical balls, all of approximately the same size (in order to make approximately constant the effects of air resistance) but of varying masses, were used to model the projectiles. The force of the trebuchet was modeled by throwing by the experimenters. The resulting projectile motion was recorded with a video camera; MotionLab analysis software was used to generate (horizontal position, vertical position, time) triplets at each frame in the trajectories and, by linear interpolation, (horizontal velocity, vertical velocity, time) triplets between each pair of consecutive frames in the trajectories. A meter stick was placed less than 5cm behind the projectiles' plane of motion for calibration of this software. The position and velocity of each projectile as functions of time were fit

by eye as parabolas and lines, respectively. The acceleration of each projectile was then taken to be the slope of the velocity fit because this was deemed more reliable than the position fit and because it was easier to quantify the error in the velocity fit.

Two trajectories were analyzed in this fashion. Due to time constraints, the results of all the lab groups were combined to yield enough data for the analysis. The other groups' procedures were similar, but the details are unknown.

Data

M (g)	a_y (m/s ²) (low)	a_y (m/s ²) (best fit)	a_y (m/s ²) (high)
48.8	9.7	10.0	10.7
51.4	9.3	9.5	11.1
57.3	9.0	10.0	10.6
75.0	9.0	9.7	10.0
141.2	9.1	9.8	10.5
148.6	9.3	9.9	10.8
165.5	9.4	10.0	10.5

Table 1: The vertical accelerations as measured by MotionLab fits of velocity and the associated masses. The uncertainty in all of the masses is 0.3g

Analysis

The accelerations in the vertical (y) direction as measured by the fits are given in Table 1 in the Data section. The accelerations in the horizontal (x) direction are not given because they are all 0. Errors were assigned to the fits by finding the maximal and minimal values of the parameters which yield apparently valid fits. A constant, the average of the “best fit” accelerations listed in Table 1, was then taken as the single parameter in a 0-degree polynomial fit to the data. The error was taken to be the standard deviation from this parameter. The fit is depicted in Figure 2.

As Figure 2 illustrates, the fit falls within the error of all the data points, so it is valid to say that this has confirmed the prediction that the vertical acceleration is constant with respect to mass.

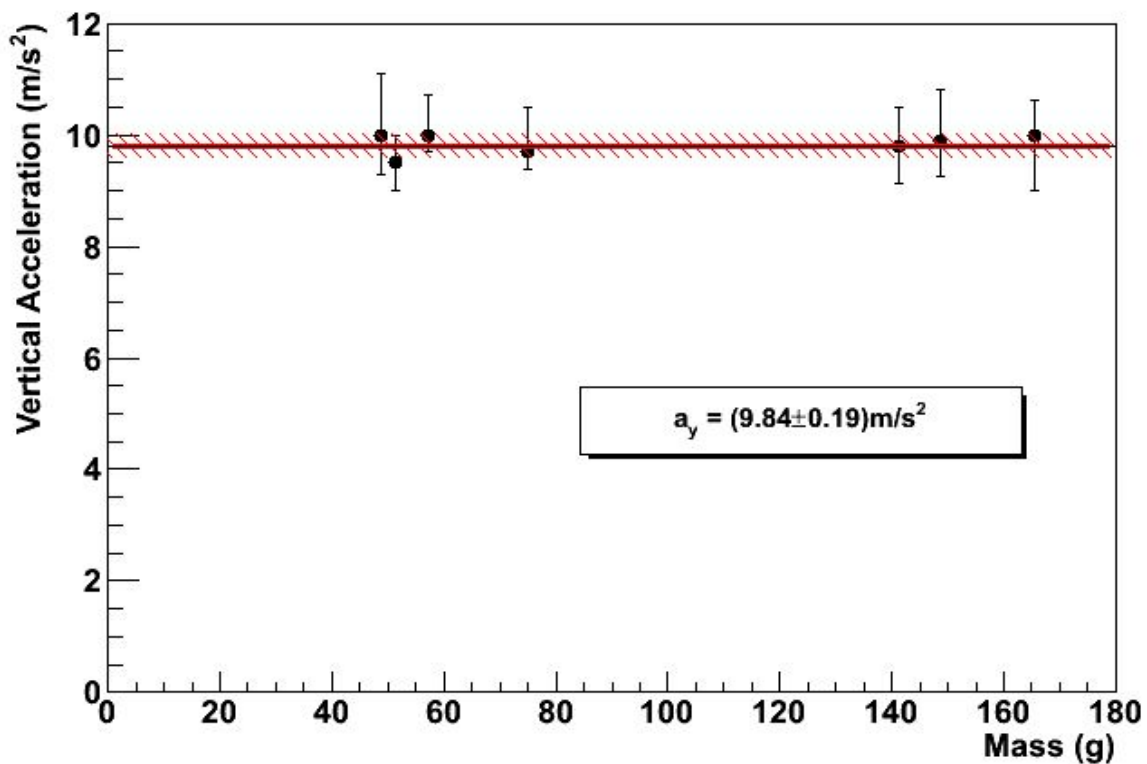


Figure 2: The measured vertical accelerations versus the respective projectile masses and the constant fit thereto. The errors in the masses are smaller than the markers.

Because all of the horizontal accelerations $a_x = 0$, the hypothesis that the horizontal acceleration of the projectiles is constant with respect to mass has been confirmed; although there exists a nonzero uncertainty in all of these measurements, 0 lies within all possible error intervals.

Possible sources of systematic error include air resistance, distortion due to the camera's optics, error in calibration due to the offset depth of the trajectories versus the meter stick, and the constraint that the first frame of the ball's motion was at time 0, which is accurate only to 0.016s. These, and any other systematics, are believed to be insignificant because the average and expected accelerations in both the horizontal and vertical directions are consistent with the individual measurements to within experimental error.

Conclusion

The motion of projectiles launched by trebuchets was modeled by thrown balls. The hypothesis that the horizontal accelerations thereof are mass-independent was confirmed in that all were measured to be 0. The hypothesis that the vertical accelerations thereof are mass-independent was confirmed in that a single, constant acceleration of 9.84m/s^2 lay within the error intervals of all of the measured data points.

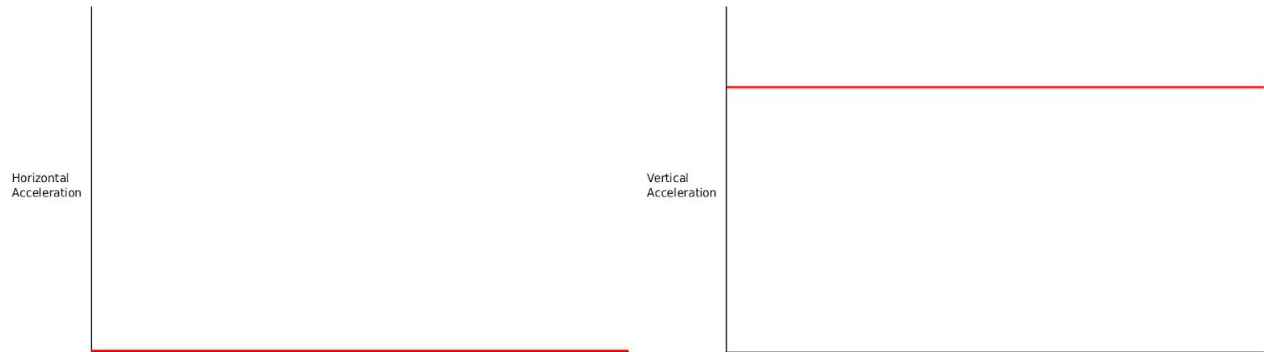
BAD SAMPLE LAB**Lab II, Problem 4**

Comte de Rochefort

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Introduction

We want to figure out how the trebuchet's projectiles will move if their mass is changed. A trebuchet is a kind of medieval catapult that uses gravity to launch rocks. First, we threw balls to simulate the rocks. We recorded them with a camera. Then, we analyzed the videos using MotionLab. Then, we decided that the acceleration does not change when the mass changes.

Prediction***Procedure***

The procedure in this experiment began with setup. We collected the following materials:

- meter stick
- tennis ball
- baseball
- video camera on tripod
- computer with MotionLab software
- stopwatch

We then positioned the camera facing the wall. We taped the meter stick to the wall.

We next recorded the videos. We threw the tennis ball in a parabolic trajectory parallel to the wall and recorded a video of it with the camera and computer. We did the same for the baseball.

We then analyzed the videos with MotionLab. We began by setting $t=0$ to the time when the ball left General Veers's hand. We then used the meter stick to calibrate the length in the video. We defined our coordinate system. It had the origin where the ball was at $t=0$, x was horizontal, and y was vertical. We then had to make predictions about the position graphs. Since there is no acceleration in the x direction, we predicted it would be a straight, linear line. Since there is acceleration in the y direction, we predicted it would be quadratic. We derived the coefficients for the predictions by measuring how high and how far the ball went with the meter stick and how long it flew with the stopwatch. The first ball flew $88 \pm 0.05\text{cm}$ in the x direction and $90 \pm 0.05\text{cm}$ in the y direction, and took $0.85 \pm 0.005\text{s}$ to complete its trajectory. The second ball flew $110 \pm 0.05\text{cm}$ in the x direction and $60 \pm 0.05\text{cm}$ in the y direction. It took $0.86 \pm 0.005\text{s}$ to complete its trajectory. The predicted equations were $x=0+1.054t$ and $y=0+4.185t-4.9t^2$ for the first ball and $x=0+0.694t$ and $y=0+4.185t-4.9t^2$ for the second ball. We then added a data point at each frame in the ball's flight. We omitted some frames near the end of the video when the ball was in the distorted region. We took 24 data points for the first ball and 29 data points for the second ball. We fit graphs to the resulting data points. The fits were $x=0+1.05t$ and $y=0+3.47t-5t^2$ for the first ball and $x=0+0.71t$ and $y=0+4.37t-5t^2$ for the second ball. We then had to predict the velocity graphs of the balls. We did this by making the t coefficient in the position function the constant in the velocity function and the t^2 coefficient in the position function the t coefficient in the velocity function. This made the xv graph a constant line and the yv graph a linear line. The predictions were $xv=1.05+0t$ and $yv=3.47-10t$ for the first ball and $xv=0.71+0t$ and $yv=4.37-10t$ for the second ball. After this, we had to fit the velocity graphs to the data points. The fits were $xv=1.05+0t$ and $yv=3.47-10t$ for the first ball and $xv=0.71+0t$ and $yv=4.37-10t$ for the second ball. The fits were the same as the predictions, so there were no errors in the predictions. We then got the accelerations from the coefficients of the fits. This was 0.5 of the t^2 coefficient in the position fit and the same as the t coefficient in the velocity fit.

After analyzing the videos, we exchanged data with the other groups, left the lab, and analyzed the data.

Data

Ball 1

mass: $57.3 \pm 0.05\text{g}$
x distance: $88 \pm 0.05\text{cm}$
y distance: $90 \pm 0.05\text{cm}$
time: $0.85 \pm 0.005\text{s}$
x prediction: $x=0+1.054t$
x fit: $x=0+1.05t$
y prediction:
 $y=0+4.185t-4.9t^2$
y fit: $y=0+3.47t-5t^2$
xv prediction: $xv=1.05+0t$
xv fit: $xv=1.05+0t$
yv prediction: $yv=3.47-10t$
yv fit: $yv=3.47-10t$

Ball 2

mass: $48.8 \pm 0.05\text{g}$
x distance: $110 \pm 0.05\text{cm}$
y distance: $60 \pm 0.05\text{cm}$
time: $0.86 \pm 0.005\text{s}$
x prediction: $x=0+0.694t$
x fit: $x=0+0.71t$
y prediction:
 $y=0+4.185t-4.9t^2$
y fit: $y=0+4.37t-5t^2$
xv prediction: $xv=0.71+0t$
xv fit: $xv=0.71+0t$
yv prediction: $yv=4.37-10t$
yv fit: $yv=4.37-10t$

Ball 3

mass: $165.5 \pm 0.05\text{g}$
x prediction: $x=0+1.126t$
x fit: $x=0+1.13t$
y prediction:
 $y=0+3.915t-4.9t^2$
y fit: $y=0+3.37t-4.9t^2$
xv prediction: $xv=1.13+0t$
xv fit: $xv=1.13+0t$
yv prediction: $yv=3.37-9.8t$
yv fit: $yv=3.37-10t$

Ball 4

mass: 51.4+/-0.05g
x prediction: $x=0+0.877t$
x fit: $x=0+0.82t$
y prediction:
 $y=0+4.469t-4.9t^2$
y fit: $y=0+3.8t-4.7t^2$
xv prediction: $xv=0.82+0t$
xv fit: $xv=0.82+0t$
yv prediction: $yv=3.8-9.4t$
yv fit: $yv=3.8-9.5t$

Ball 5

mass: 141.2+/-0.05g
x prediction: $x=0+1.203t$
x fit: $x=0+1.21t$
y prediction:
 $y=0+3.258t-4.9t^2$
y fit: $y=0+3.1t-4.9t^2$
xv prediction: $xv=1.21+0t$
xv fit: $xv=1.21+0t$
yv prediction: $yv=3.1-9.8t$
yv fit: $yv=3.1-9.8t$

Ball 6

mass: 148.6+/-0.05g
x prediction: $x=0+1.281t$
x fit: $x=0+1.4t$
y prediction:
 $y=0+3.258t-4.9t^2$
y fit: $y=0+4.1t-4.95t^2$
xv prediction: $xv=1.4+0t$
xv fit: $xv=1.4+0t$
yv prediction: $yv=4.1-9.9t$
yv fit: $yv=4.1-9.9t$

Ball 7

mass: 75.0+/-0.05g
x prediction: $x=0+0.943t$
x fit: $x=0+1.07t$
y prediction:
 $y=0+3.895t-4.9t^2$
y fit: $y=0+3.3t-4.85t^2$
xv prediction: $xv=1.07+0t$
xv fit: $xv=1.07+0t$
yv prediction: $yv=3.3-9.7t$
yv fit: $yv=3.3-9.7t$

Analysis

We calculate the accelerations from the fits because we know $x = x_0 + v_0t + \frac{1}{2}at^2$. All the accelerations in the x direction are therefore 0. The accelerations in the y direction are -10m/s^2 , -10m/s^2 , -9.8m/s^2 , -9.4m/s^2 , -9.8m/s^2 , -9.9m/s^2 , -9.7m/s^2 .

We know that the x accelerations should be 0 because we are ignoring air resistance. We know that the y accelerations should be -9.8m/s^2 . All of the y accelerations are close to this. They differ by 0.2m/s^2 , 0.2m/s^2 , 0m/s^2 , 4m/s^2 , 0m/s^2 , 0.1m/s^2 , and 0.1m/s^2 ; these are all small.

There are several important sources of error in this lab. One is the fisheye effect of the camera lens. Another is the finite accuracy of the measuring devices. The stopwatch can only measure to 0.01s, and the meter stick can only measure to 0.001m, so these measurements are only accurate to half of those values. There is error in MotionLab, too, as can be seen in the differences between some of the position and velocity fits. There was error in that we couldn't throw the balls exactly the same every time. Finally, there could have been human error. We know that all of these errors were not significant, though, because all of the measurements of acceleration were so close to the known right values.

Conclusion

We measured the acceleration of seven balls in projectile motion and got things very close to the right values every time. We can therefore say that the mass dependence of the accelerations in the x and y directions are both constant. In the x direction, it is 0m/s^2 , and in the y direction, it is -9.8m/s^2 . This was true for all the masses. This is the same as our original prediction. We can therefore say that this experiment was a success.